

```

...
G00 X4. Y2. ;
G01 F20.0 Z-0.1 ;
G03 F20.0 I-2.0 J0. X0. Y2. ;
...
M30 ;
%
```

Method 2:

```

%
T01 M06 ;
...
G00 X4. Y2. ;
G01 F20.0 Z-0.1 ;
G03 F20.0 X0. Y2. R2. ;
...M30 ;
%
```

Below is an example of how to cut a 2" (or 2 mm) radius circle:

```

%
T01 M06 ;
...
G00 X4. Y2. ;
G01 F20.0 Z-0.1 ;
G02 F20.0 I2.0 J0. ;
...
M30 ;
%
```

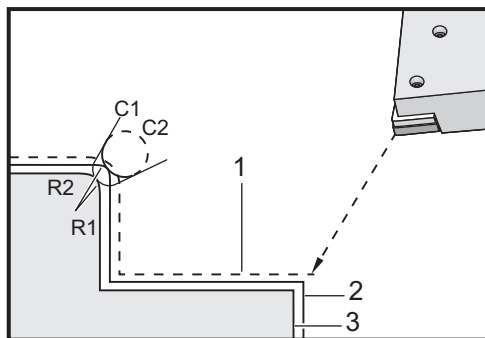
5.7 Tool Nose Compensation

Tool Nose Compensation (TNC) is a feature that lets you adjust a programmed tool path in for different cutter sizes, or for normal cutter wear. With TNC. you only need to enter minimal offset data when you run a program. You do not need to do additional programming.

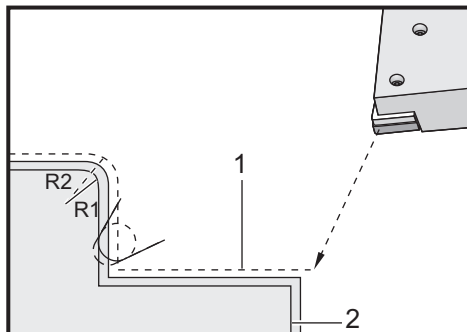
5.7.1 Tool Nose Compensation - Programming

Tool Nose Compensation is used when the tool nose radius changes, and cutter wear is to be accounted for with curved surfaces or tapered cuts. Tool Nose Compensation generally does not need to be used when programmed cuts are solely along the X- or Z-axis. For taper and circular cuts, as the tool nose radius changes, under or overcutting can occur. In the figure, suppose that immediately after setup, C1 is the radius of the cutter that cuts the programmed tool path. As the cutter wears to C2, the operator might adjust the tool geometry offset to bring the part length and diameter to dimension. If this were done, a smaller radius would occur. If tool nose compensation is used, a correct cut is achieved. The control automatically adjusts the programmed path based on the offset for tool nose radius as set up in the control. The control alters or generates code to cut the proper part geometry.

F5.7: Cutting path without tool nose compensation: [1] Tool Path, [2] Cut after wear [3] Desired cut.



F5.8: Cutting path with tool nose compensation: [1] Compensated tool path, [2] Desired cut and programmed tool path.



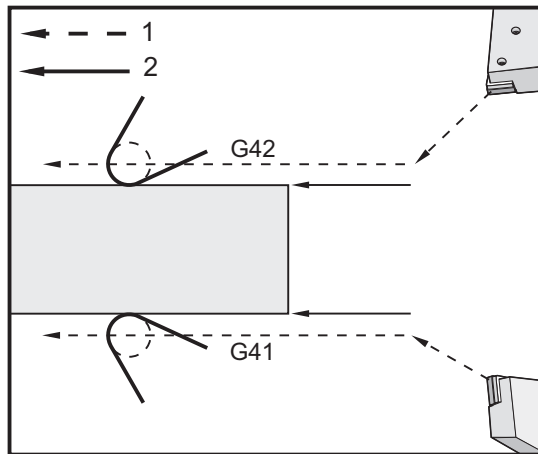
**NOTE:**

The second programmed path coincides with the final part dimension. Although parts do not have to be programmed using tool nose compensation, it is the preferred method because it makes program problems easier to detect and resolve.

5.7.2 Tool Nose Compensation Concept

Tool nose compensation works by shifting the Programmed Tool Path to the right or to the left. The programmer usually programs the tool path to the finished size. When tool nose compensation is used, the control compensates for the radius of the tool based on special instructions written into the program. Two G-code commands are used to do this for compensation within a two-dimensional plane. G41 commands the control to shift to the left of the programmed tool path, and G42 commands the control to shift to the right of the programmed tool path. Another command, G40, is provided to cancel any shift made by tool nose compensation.

F5.9: TNC Shift Direction: [1] Toolpath relative to the workpiece, [2] Programmed toolpath.



The shift direction is based on the direction of the tool movement relative to the tool and which side of the part it is on. When thinking about which direction the compensated shift occurs in tool nose compensation, imagine looking down the tool tip and steering the tool. Commanding G41 moves the tool tip to the left and G42 moves the tool tip to the right. This means that normal O.D. turning requires G42 for correct tool compensation, while normal I.D. turning requires G41.